

Arena Companion

A mobile field assistant for volunteers working in and around event arenas. The product is designed to make the next action obvious, reduce uncertainty, and keep operational information useful even in weak network conditions.

PRIMARY GOAL

Reduce uncertainty during live volunteer work

MAIN USERS

Volunteers and shift leaders

PLATFORM

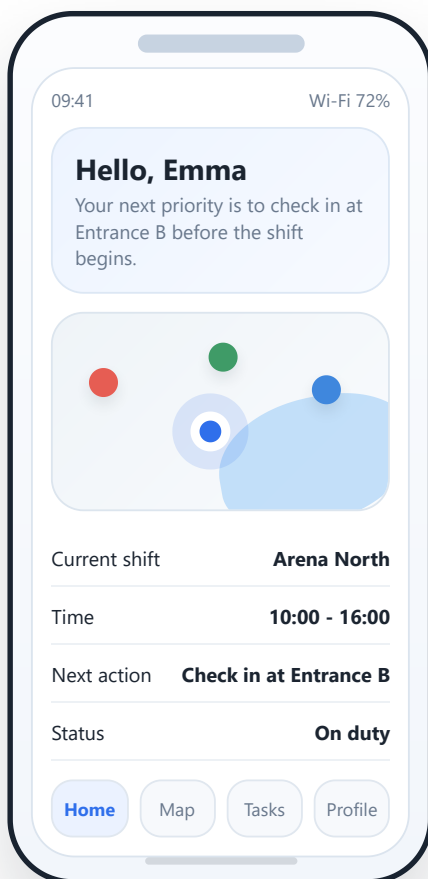
Android mobile application

CORE LENS

UI should help action, not administration

HOME SCREEN CONCEPT

Main screen with annotated product meaning

**1**

Current shift is visible immediately

The user should not need to search for time, role, or assigned arena.

2

Map supports fast orientation

The top area gives instant spatial context instead of long text instructions.

3

Next action is explicit

The app reduces uncertainty by surfacing the single most important thing to do next.

4

Navigation stays small

Only four main tabs are needed for the MVP: Home, Map, Tasks, and Profile.

USER AND CONTEXT

Who this product serves

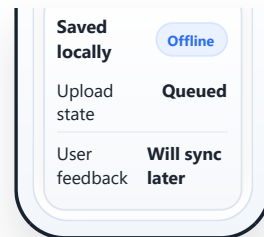
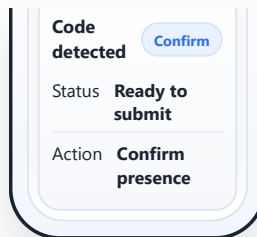
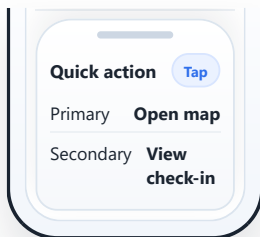
Volunteers need quick orientation and clear tasks

Shift leaders need structured check-ins and issue feedback

Arena staff influence live operational conditions

INTERACTIVE WALKTHROUGH

What using the product feels like during one shift



WHY THE FLOW WORKS

Three interaction principles

Short path to action

The app should help the user act with one or two taps, not browse through menus.

Clear confirmation

Check-in and reporting screens must always communicate whether the action succeeded.

Strong feedback

Weak connectivity is expected, so the interface must explain local save and sync status clearly.

Field-friendly structure

The user should always know the next task, current place, and reporting path.

MVP FEATURE SET

What the first usable version must include

Shift overview

Time, role, assigned area, and immediate instructions.

OpenStreetMap view

Venue orientation with checkpoints, incidents, and current context.

QR check-in

Fast presence confirmation at checkpoints or task locations.

Incident report

Structured issue capture with location, category, and optional photo.

SYSTEM FEEDBACK

What the interface should always communicate

Action started

The user can tell that the selected task or report flow has opened correctly.

Confirmation ready

Check-in states should clearly indicate when the app is ready for final confirmation.

Saved offline

Weak signal should never make the user wonder whether the report was lost.

Synced later

Background upload should resolve queued items without creating extra admin work.

DATA MODEL

Core entities the app depends on

Volunteer

Role

Shift

Arena

Zone

Task

Check-in

Incident

Message

Location

Layer

Stores

Why

Backend

Identity, shifts, tasks, incidents, messages

Provides the shared operational source of truth.

Local device

Cached shifts, map state, queued reports, recent tasks

Maintains usability when connectivity is weak.

TECHNOLOGY STACK

Recommended implementation path

Kotlin

Jetpack Compose

MVVM

Room

WorkManager

DataStore

Retrofit

OpenStreetMap

osmdroid

Android Location

ML Kit / ZXing

Supabase

PostgreSQL

This keeps the product Android-native, avoids a Google Maps dependency, and still supports real authentication, structured data, QR flows, and offline behavior.

SYSTEM STRUCTURE

How the mobile UI connects to the wider system

Mobile client

Compose UI, task flows, map view, QR scan, incident form, local cache, and session state.



Backend services

Authentication, volunteer identity, shifts, tasks, reports, messages, and media storage.

Offline layer

Stores a working subset of data and pending actions for unstable network conditions.



Background sync

Uploads queued actions later and keeps the user informed about local save vs synced state.

AUTHENTICATION

Login and session handling

Proposed flow

- Volunteer signs in through backend authentication.
- The app stores a session token locally instead of a raw password.
- Shift data and recent tasks are cached after login for faster reuse.
- Session state remains visible even when the network becomes unstable.

CONSTRAINTS

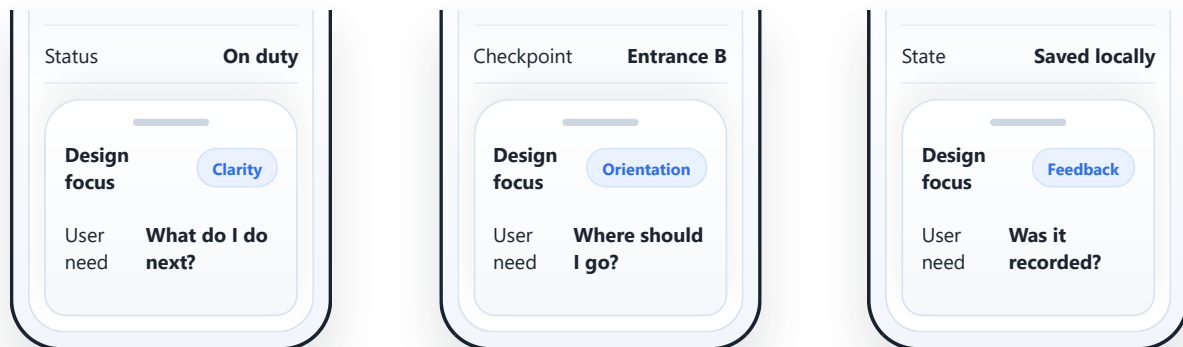
Technical realities already considered

Design implications

- No Google Maps dependency: OpenStreetMap-based rendering is preferred.
- Battery and permission usage must stay reasonable during long shifts.
- Offline-first handling matters because crowd-heavy venues weaken connectivity.
- Camera, QR, and location features should support the task instead of slowing it down.

MVP SCOPE

The first complete volunteer journey



SCOPE DECISIONS

What belongs in the first milestone

Core MVP boundary

- Show the current shift, role, and next action immediately.
- Provide map-based orientation with OpenStreetMap support.
- Allow QR check-in and a short structured reporting flow.
- Preserve critical actions locally when the network is weak.

OPEN DESIGN QUESTIONS

What still needs product decisions

- How much map detail is helpful before it becomes distracting?
- Which incidents need the shortest reporting flow?
- Should shift leaders use the same client or a lighter variant?
- How much offline support is enough for the first usable prototype?

ROADMAP

Product evolution roadmap

